

High-Field Angular I_c Metrology for REBCO Tapes: Combined Pulsed-Current and Rotation Techniques

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Abstract—Understanding flux pinning and its angular dependence in REBCO coated conductors is essential for both vortex physics and next-generation high-field applications. However, conventional characterization methods suffer from inherent limitations: full-width tape measurements capture the total transport critical current I_c but provide limited angular dependence, whereas microbridge measurements offer detailed angular dependence but may deviate from intrinsic performance of practical conductors. In this study, we developed a hybrid metrology that combines pulsed high-current measurement at fixed angles and rotation-based DC techniques to achieve comprehensive $I_c(B, T, \theta)$ characterization up to 24 T. The pulsed-current system supplies currents up to 5 kA for full-width tapes, while the rotational DC setup enables precise angular measurement on microbridges with currents up to 20 A. By using identical sample batches, we demonstrated that these two approaches provided complementary insights into high-field transport behavior. We further found that trapped magnetic fields around microbridges significantly influence the I_c at low fields, emphasizing the importance of local field homogeneity for accurate flux-pinning evaluation. This combined methodology establishes a robust framework for assessing angular-dependent flux pinning under very high magnetic fields and provides valuable guidance for superconducting magnet design, materials optimization, and fundamental studies on vortex matter under extreme conditions.

Index Terms—Critical current, flux pinning, high magnetic field, pulsed current, REBCO coated conductor.

I. INTRODUCTION

HIGH-TEMPERATURE superconductors based on $\text{REBa}_2\text{Cu}_3\text{O}_{7-\delta}$ (REBCO, RE = rare earth) have enabled

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the realization of superconducting magnets generating magnetic fields beyond 30 T [1], [2]. Such achievements have opened new frontiers in high-field science, including unconventional quantum phenomena such as superconductivity in UTe_2 [3], high-resolution nuclear magnetic resonance [4], [5], compact fusion power plants [6], [7], [8], and accelerators for high-energy physics. In recent years, several national programs have launched 40 T-class magnet projects [9], and our group has also initiated the development of 33 T and 40 T class magnets using REBCO insert coils [2], [10], [11].

For the design of these high-field magnets, precise knowledge of the transport critical current, $I_c(B, T, \theta)$ —the dependence on magnetic field magnitude, temperature, and field angle—is indispensable [12], [13]. Owing to the intrinsic crystalline anisotropy of REBCO, the angular dependence of I_c reflects both the anisotropic condensation energy and the flux-pinning landscape, thereby directly governing conductor performance under complex magnetic field configurations [14], [15]. Since the early 2000s, artificial pinning centers have been actively introduced to enhance performance, making it increasingly important to analyze flux-pinning anisotropy in detail [16], [17], [18]. The flux-pinning landscape arises from the interplay among vortex–vortex interactions, vortex–defect interactions, and thermal fluctuations, all of which depend sensitively on magnetic field, temperature, and orientation—rendering this system an intriguing platform for exploring vortex phase diagrams and pinning mechanisms [19].

In recent years, several groups have developed advanced techniques for high-field I_c characterization such as magnetization torque-based angular studies [20], [21], kiloampere-class transport measurements with fine temperature stabilization and *ex-situ* angular control [22], [23], [24], [25]. However, REBCO conductors now exhibit kiloampere-class performance even at low temperatures, which makes transport measurements increasingly challenging in cryogenic environments with limited cooling power. To date, no measurement technique has simultaneously achieved both kiloampere-level current capability [26], [27], [28] and *in-situ* angular control under high magnetic fields [29].

To overcome these limitations, we developed a hybrid measurement framework combining pulsed high-current tests for full-width REBCO tapes with *in-situ* rotation DC tests for micro-patterned bridges. This approach enables direct comparison between full-width and microbridge samples under identical field and temperature conditions. Furthermore, the microbridge geometry was systematically optimized to minimize the

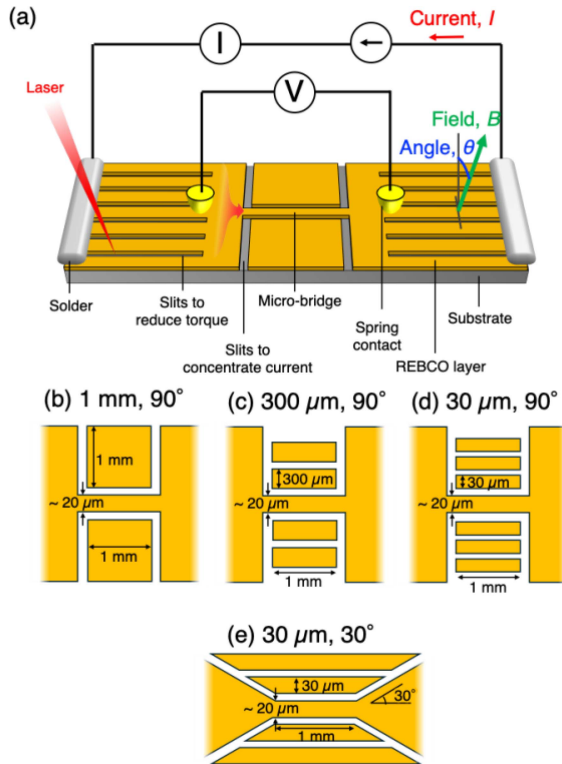


Fig. 1. Schematic drawing of (a) overview of microbridge and (b-e) 4 detailed patterns. All microbridges have a length of 1 mm and a width of $\sim 25 \mu\text{m}$.

influence of trapped magnetic fields. This study therefore establishes a unified experimental basis for evaluating angular-dependent $I_c(B, T, \theta)$ behavior and anisotropic high-field flux-pinning characteristics in REBCO conductors.

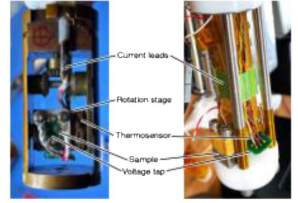
II. EXPERIMENTAL METHOD

A. Sample and Micro-Patterning

Commercial REBCO coated conductors (Fujikura, FESC-SCH04(40), 4 mm wide) were used, where “(40)” indicates the presence of $40 \mu\text{m}$ -thick Cu stabilizer layers on each side of the tape. These tapes, provided in late 2020 to early 2021, originated from the same batch used for the large prototype coil of the 33 T cryogen-free superconducting magnet, ensuring consistency with practical magnet materials [30]. The 4 mm-wide tapes exhibited a critical current of 212–220 A at 77 K under self-field. The superconducting layer consisted of BaHfO_3 -doped $\text{EuBa}_2\text{Cu}_3\text{O}_y$ with a thickness of approximately $2.4 \mu\text{m}$. For pulsed-current measurements, full-width (4 mm) samples were used without additional patterning, and the tape ends were trimmed to a total length of 32 mm.

To limit the transport current and reduce Joule heating during the measurement, microbridges were fabricated from the same tape using laser-based patterning. Fig. 1 shows the schematic drawing of the sample and four microbridge designs including electrical circuit, each with a bridge length of 1 mm and a width of approximately $25 \mu\text{m}$. The surrounding geometry of the bridge region was varied to investigate the influence of trapped magnetic fields near the bridges. In addition, the current-pads

TABLE I
SPECIFICATIONS OF THE PULSED-CURRENT AND ROTATION-BASED DC MEASUREMENT SYSTEMS

Probe	DC-rotation	Pulse-fixed
Maximum current	20 A (typ.)	5 kA (typ.)
Current type	DC	Pulsed ($\sim\text{ms}$)
Temperature	Variable, 4.2–77 K	
Magnetic field	up to 25 T	
Angular control	Continuous rotation	Fixed ($B \perp$ tape)
Sample size (mm^2)	4×10	4×32
Connection for current	Clamp or solder	Clamp
Connection for voltage	Spring or press contact	Press contact
Outlook		

were designed with slits to minimize magnetic flux trapping and magnetization, thereby reducing torque during sample rotation. The laser patterning was performed at room temperature, where scanning laser irradiation was applied to the sample to remove the REBCO layer together with the Cu and Ag stabilizer layer. Optical microscopy confirmed that the laser patterning completely penetrated through the superconducting and stabilizer layers, reaching the substrate. At 77 K under self-field, the microbridge exhibited a critical current of about 1.4 A.

B. Hybrid Measurement Systems for High-Field I_c Characterization

Two complementary measurement systems were employed to evaluate the transport critical current under high magnetic fields: a pulsed-current system for high-current full-width tape measurements and a rotation-based DC system for low-current microbridge measurements. Table I summarizes the principal specifications of both systems.

The pulsed-current setup can deliver currents up to 5 kA with pulse durations of several milliseconds. The current waveform contains a plateau region during which the voltage was recorded under a nearly constant current. The power-supply circuit has been described in detail in a previous publication [28]. Inductive noise was minimized by suppressing probe vibration and reducing magnetic flux linkage in the voltage leads.

For angular-dependence measurements, a cryogenic rotation probe was operated in a DC configuration [15]. The probe allows continuous rotation around the tape’s longitudinal axis with an angular resolution better than 0.1° , driven by a stepping motor located at room temperature and connected to a gear assembly that transmits motion to the sample stage inside the cryostat. Low-resistance soldered contacts for current pads

ensured thermal stability and mechanical robustness during rotation. The microbridge sample was mechanically clamped by a fiber-reinforced-plastic plate integrating spring-type voltage contacts, effectively preventing delamination during repeated rotation.

Both systems share a common cryogenic platform and magnetic-field configuration, enabling direct comparison between the full-width and microbridge samples. Measurements were performed in the 25 T cryogen-free superconducting magnet (25T-CSM [31], [32]) up to 24 T, with cooling provided by a helium gas-flow cryostat. The temperature range was 4.2–77 K.

In both measurement systems, I_c was determined from transport current–voltage (I – V) characteristics using the same electric-field criterion of $1 \mu\text{V cm}^{-1}$. In the DC rotation measurement system, the transport current was incrementally changed and the voltage was measured at each constant-current step. For angular-dependence measurements, the magnetic field and temperature were fixed and I_c was evaluated at each field angle, repeating the I_c measurement while systematically changing the angle. For field-dependence measurements, the temperature and field angle were fixed, and I_c measurements were performed at each stabilized magnetic field. In the pulsed-current system, a current pulse with a well-defined plateau was applied, and the voltage was measured during the constant-current plateau, allowing quasi-steady-state I – V measurements.

III. RESULTS AND DISCUSSION

A. Optimization of Microbridge Pattern

Fig. 2(a) shows the normalized critical current $I_c/I_c^{\max}$ of the sample as shown Fig. 1(b) with square pattern as a function of magnetic field at 4.2 K, 20 K, and 77 K, where $I_c^{\max}$ is defined as the maximum critical current. A clear hysteresis was observed between the field-up and field-down sweeps, indicating the presence of trapped magnetic flux within the sample as shown in the inset of the Fig. 2(a). During the field-down process at 4.2 K, a distinct peak appears around 0.6 T, while at higher temperatures the peak shifts to lower fields—approximately 0.3 T at 20 K and 0.04 T at 77 K. The proportional decrease of the peak field with increasing temperature follows the temperature dependence of I_c at self-fields: I_c at self-fields at 77 K and 20 K differs by roughly one order of magnitude, and that at 4.2 K is about 1.5 times larger. This correlation implies that the local magnetic field is influenced by trapped field, similar to behavior observed in weak-link systems such as bicrystal or polycrystalline bulk superconductors [33], [34], [35], [36]. According to Brandt's model [37], the magnitude of the trapped field in the thin superconductor scales as the product of the critical current density J_c and thickness. Using $J_c \sim 30 \text{ MA/cm}^2$ at 4.2 K under self-field [26], the estimated trapped field is approximately 0.7 T, which agrees well with the observed peak value. At higher magnetic fields, the I_c values obtained during field-up and field-down sweeps converge, consistent with the reduction of trapped flux as the applied field increases. Above 6 T at 20 K and 8 T at 4.2 K, the difference in I_c between the two processes becomes less than 3%.

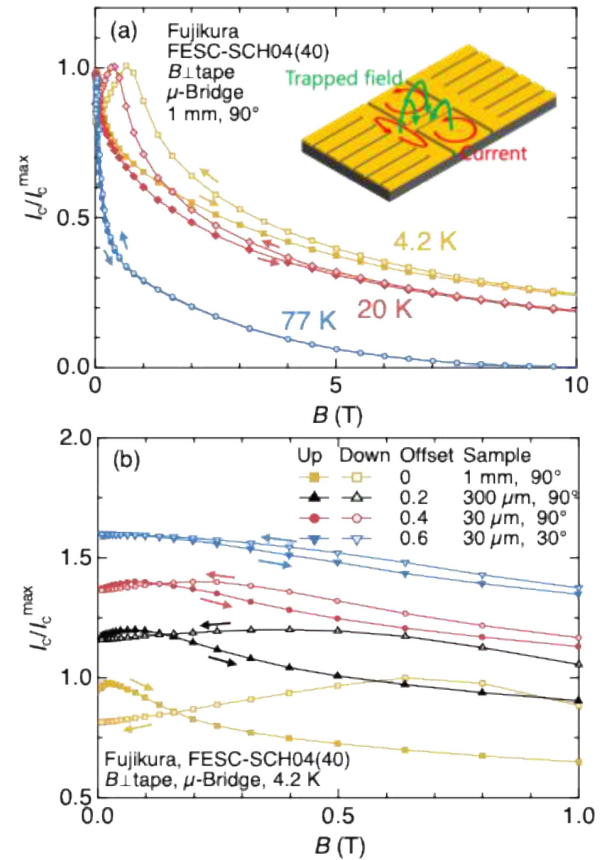


Fig. 2. Field dependence of the normalized I_c for (a) the square pattern measured at 4.2 K, 20 K, and 77 K, (b) the various microbridge patterns measured at 4.2 K.

Fig. 2(b) compares the magnetic hysteresis of normalized I_c at 4.2 K for various microbridge geometries and current-pad configurations. The data are displayed with offsets for clarity. Introducing slits around the bridge to reduce shielding currents and narrowing the surrounding conductor region from 1 mm to 300 μm decreased the peak field from 0.6 T to 0.4 T. Further reducing the slit spacing to 30 μm and modifying the current-lead opening angle suppressed the residual peak to 0.1 T and eventually to nearly 0 T. These results indicate that not only the bridge geometry but also the magnetic environment around the current pads significantly affects the self-field I_c . Consequently, the pattern shown in Fig. 1(e) was identified as the adequate design for precise angular-dependent $I_c(B, T, \theta)$ measurements under high magnetic fields.

B. Full-Width and Microbridge Samples: Flux Pinning Behavior

Fig. 3 shows the comparison between the full-width (FW) and microbridge (μBr) samples to verify the consistency of the measurement systems. The data for FW in Fig. 3 are identical to those reported in our previous study [26]. The I_c of the μBr sample was scaled to an equivalent 4 mm width using the ratio of the self-field I_c at 77 K: the microbridge corresponds to an effective width of 25.6 μm , and a scaling factor of 156 was

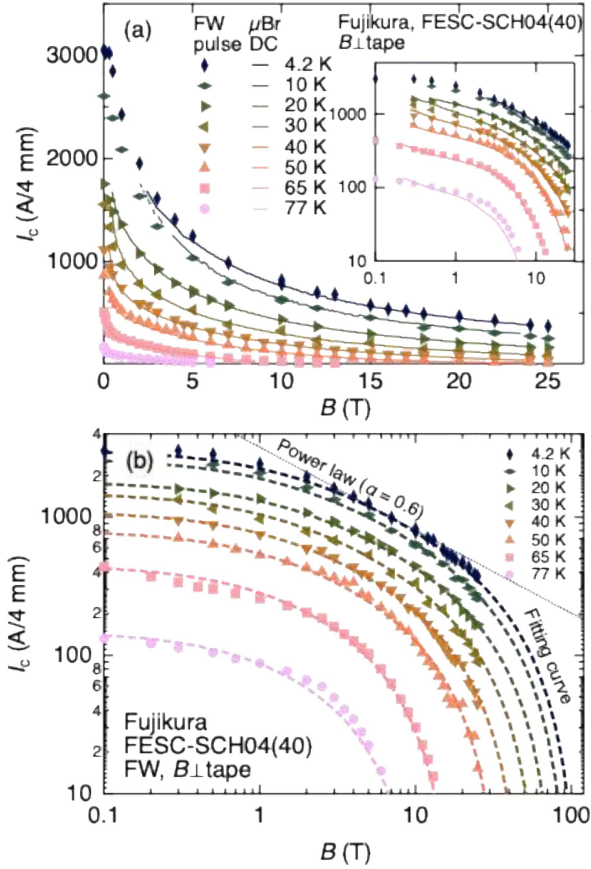


Fig. 3. (a) Field dependence of the I_c for Fujikura FESC-SCH04(40) tape with full-width (FW) and microbridge (μ Br) at 4.2–77 K. The inset shows the same data on a double-logarithmic scale. (b) Field dependence of the I_c with fitting results.

applied. In the high-field region, the two datasets agreed within $\pm 5\%$, confirming that laser micro-patterning did not introduce any measurable degradation. In contrast, below 3 T, the μ Br I_c was up to 8% higher than the FW I_c . This enhancement can be attributed to the larger self-field generated in the FW tape, which suppresses the apparent I_c . As discussed in Section A, the microbridge geometry can generate a local self-field of about 0.6 T at 4.2 K, consistent with this estimation.

Fig. 3(b) analyzes the effective flux-pinning behavior in the REBCO coated conductor. The $I_c(B)$ characteristics follow a simple power-law dependence $I_c(B) \propto B^{-\alpha}$ in some cases [19], [24], [38], typical of collective pinning. However, at high fields, the flux-line lattice experiences plastic deformation [39], and I_c rapidly decreases once the vortex lattice softens. At low temperatures (e.g., 4.2 K and 20 K), assuming a simple power-law leads to an overestimation of I_c in the high-field regime such as over 20 T. To account for this, a generalized scaling law based on the p - q form was employed,

$$I_c(B) = I_c(0) \left(1 + \frac{B}{B_0}\right)^{-p} \left(1 - \frac{B}{B_{irr}}\right)^q \quad (1)$$

where B_0 is the accommodation field, B_{irr} is the irreversible field, p and q are the dimensionless parameter. The first term becomes the Bean model [40] or the Kim model [41] for $p = 0$

TABLE II
FITTING PARAMETERS FOR THE FIELD DEPENDENCE OF I_c FOR REBCO TAPE (FUJIKURA, FESC-SCH04(40))

T (K)	I_{c0} (A)	B_0 (T)	B_{irr} (T)	p	q
4.2	3050	1.6	117	0.6	2
10	2650	1.4	102	0.58	2
20	1800	1.5	84	0.58	2
30	1500	1.3	67	0.58	2
40	1100	1.3	52	0.58	2
50	800	1.2	38	0.55	2
65	450	1.1	19	0.55	2
77	150	0.9	12	0.5	2

or 1, respectively. Empirically, p is in a range of 0.5–0.6 in the collective vortex pinning regime [19]. The last multiplied term of the formula is described by the empirical Kramer model where q is ~ 2 in the low temperature superconducting materials [42]. Empirically, q is ranged at 2.0–3.0 for the REBCO films and bulks. In this study, we assume p and q as in ranges of 0.5–0.6 and 2.0–3.0 for the fitting, respectively. As results, p was close to 0.6 and q was 2 for all conditions. Table II shows the fitting parameters for each temperature. The temperature dependence irreversibility field B_{irr} was estimated using

$$B_{irr}(T) = B_{irr}(0) \left(\frac{T}{T_c}\right)^{-a} \left(1 - \frac{T}{T_c}\right)^b \quad (2)$$

where a and b are the dimensionless parameters, $B_{irr}(0)$ is the irreversible magnetic field at 0 K. We assume $B_{irr}(0) = 120$ T according to the REBCO films with using the pulsed magnetic field [43]. For the parameters a and b , we refer to the experimental results for the REBCO films under static magnetic fields up to 35 T [44]. Then, a and b is fixed as 0.015 and 1.42, respectively [43], [44], [45].

The excellent agreement of the fitted curves across all fields and temperatures indicates the validity of scaling framework, while multiple pinning mechanisms contribute with varying relative importance. In the similar system of $\text{EuBa}_2\text{Cu}_3\text{O}_y$ with BaHfO_3 doping, the artificial pinning centers consist of inclined and discontinuous nanorods rather than continuous c -axis-correlated columns [46], [47]. These structural features suggest that the BaHfO_3 phase act as random pinning centers, consistent with the field-scaling behavior observed here.

C. Angular Dependence and Scaling Analysis

Fig. 4 shows the angular dependence of the critical current $I_c(\theta)$ for the REBCO tape, measured up to 24 T and over the temperature range 4.2–77 K. Below 50 K, $I_c(\theta)$ reaches its maximum at $\theta = 90^\circ$ for all magnetic fields, corresponding to strong pinning by intrinsic pinning at low temperatures or by stacking faults at higher temperatures. The peak becomes sharper as both temperature and field increase, reflecting a gradual transition in the dominant pinning mechanism from intrinsic pinning to stacking-fault pinning. At low temperatures (below 30 K), $I_c(\theta)$

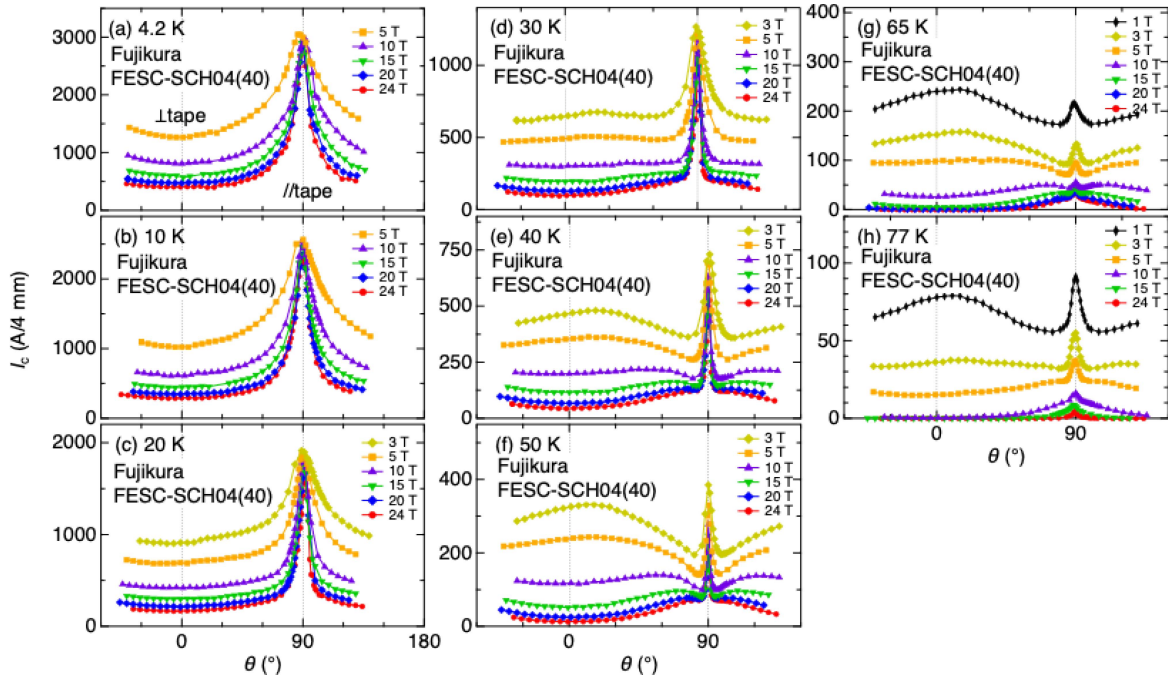


Fig. 4. Field angular dependence of the I_c for the Fujikura FESC-SCH04(40) REBCO tape measured at various temperatures and magnetic fields. $\theta = 0^\circ$ is defined as the magnetic fields perpendicular to the tape.

shows a minimum at $\theta = 0^\circ$. Above 40 K, a weak peak appears around $\theta = 0^\circ$ under fields below 5 T, suggesting the emergence of weakly c -axis correlated pinning. This behavior indicates that partially aligned BaHfO₃ nanorods act as pinning centers, providing moderate c -axis correlation. In the intermediate temperature range (40–65 K), a shoulder develops near $\theta = 80^\circ$. This feature likely arises from the reduced pinning efficiency when obliquely oriented, discontinuous nanorods overlap less with the magnetic-flux lines as the field approaches the ab-plane.

These results suggest that at low temperatures, random weak pins and intrinsic pinning dominate; at intermediate temperatures, partially correlated nanorod pinning contributes; and at high temperatures, even these nanorod effects become weak. A more detailed analysis would require decomposition of the correlated pinning contribution [15] or interpretation based on entropy-driven vortex models [25], [48].

To facilitate its use in magnet design, this field-angle dependence can be simplified using the $B\cos\theta$ scaling at low temperatures [46], [49]. Fig. 5 shows the field-angle scaling behavior of the critical current I_c for the Fujikura FESC-SCH04(40) REBCO tape. Fig. 5(a) presents the field dependence $I_c(B\cos\theta)$ at 4.2 K and 20 K, calculated from the angular data $I_c(\theta)$. The horizontal axis represents the perpendicular field component $B\cos\theta$. At 4.2 K, the $I_c(B\cos\theta)$ curves at different magnetic fields collapse onto a single trend, confirming that the c -axis component of the magnetic field primarily governs flux motion in this regime. This behavior, previously observed in YBCO films without artificial pinning centers, indicates the coexistence of strong intrinsic pinning and weak random pinning [49]. The I_c values for $\theta = 0^\circ$ and the fitting curves obtained from Fig. 3(b) are also plotted for comparison. At 20 K, however, the scaling no longer

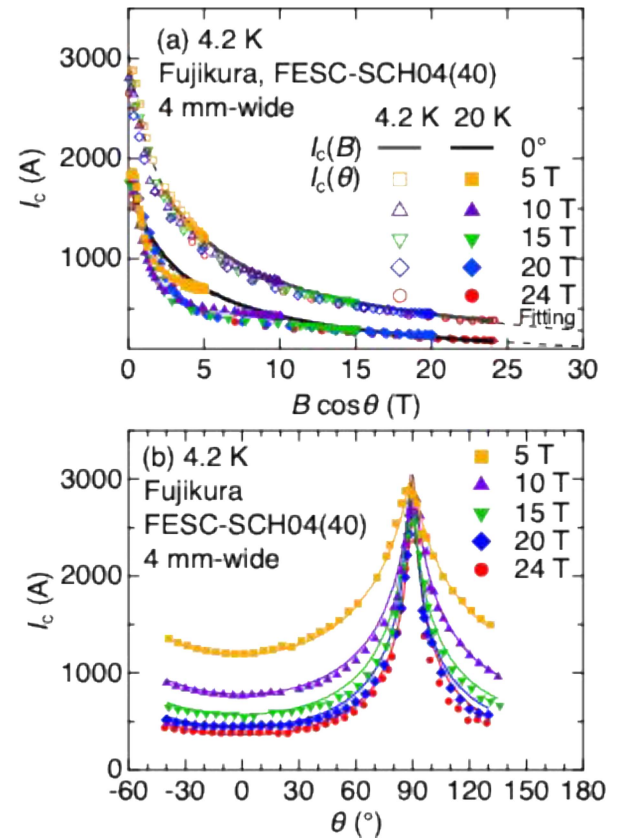


Fig. 5. (a) $B\cos\theta$ scaling of the I_c for Fujikura FESC-SCH04(40) at 4.2 and 20 K. (b) Field angular dependence of the I_c with fitting results by $B\cos\theta$ scaling.

holds. This deviation reflects the increasing influence of artificial pinning centers, which enhance I_c by up to 40% compared with the intrinsic scaling. Therefore, the application of simple $B\cos\theta$ scaling to estimate I_c at higher temperatures must be treated with caution.

Fig. 5(b) again plots the angular dependence $I_c(B, \theta)$, formulated as

$$I_c^{4.2\text{ K}}(B, \theta) = 3050 \left(1 + \frac{B \cos \theta}{1.6}\right)^{-0.6} \left(1 - \frac{B \cos \theta}{117}\right)^2 \quad (3)$$

The experimental data are well reproduced by the fitting based on the collective flux-pinning model, with a remaining overestimation of about 8% at the intermediate angle. This deviation likely arises from the coexistence of multiple pinning mechanisms that cannot be fully captured by a single scaling parameter. Nevertheless, for this REBCO conductor, the fitting approach provides a sufficiently reliable approximation for use in magnet design.

IV. CONCLUSION

A hybrid high-field $I_c(B, T, \theta)$ measurement framework combining pulsed-current and DC rotation techniques was developed to characterize REBCO conductors over a wide range of currents, magnetic fields, field orientation, and temperatures. Using identical Fujikura FESC-SCH04(40) tapes as employed in practical high-field magnets, consistent results between the pulsed and DC systems confirmed the reliability of both the measurement approach and the laser micro-patterning process.

Optimization of the microbridge geometry effectively reduced trapped-field effects and improved measurement stability up to 25 T. The field dependence of $I_c(B)$ was successfully described by a generalized scaling law, revealing a crossover from collective to random pinning at high fields. Angular-dependent measurements further clarified the evolution of the pinning landscape: intrinsic and weak random pinning dominate at low temperatures, partially correlated BaHfO₃ nanorods contribute in the intermediate range, and their effect weakens near 77 K.

At 4.2 K, $B\cos\theta$ scaling was found to hold accurately, enabling a simplified estimation of the effective current limit based on the perpendicular magnetic field component. This simplification provides a practical framework for preliminary magnet design and current-sharing simulations, while at higher temperatures or in strong in-plane fields, the full $I_c(B, \theta)$ dependence must be employed for precise modeling.

This study establishes a robust experimental foundation for quantitative evaluation of flux-pinning anisotropy and provides essential design guidelines for next-generation REBCO-based high-field magnets.

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